

LEVEL-VI

SINGLE ANSWER QUESTIONS

1. If the three points with position vectors $(1, a, b)$, $(a, 2, b)$ and $(a, b, 3)$ are collinear in space, then the value of $a + b$ is

(A) 3 (B) 4 (C) 5 (D) none

2. Consider the three points P, Q, R whose coordinates are respectively $(2, 5, -4)$, $(1, 4, -3)$, $(4, 7, -6)$ then the ratio in which the point Q divides PR.

(A) 1 : 3 (B) 1 : 2 (C) -1 : 3 (D) -1 : 2

3. Let $\vec{r} = \vec{a} + \lambda \vec{\ell}$ and $\vec{r} = \vec{b} + \mu \vec{m}$ be two lines in space where

$$\vec{a} = 5\hat{i} + \hat{j} + 2\hat{k}, \vec{b} = -\hat{i} + 7\hat{j} + 8\hat{k}, \vec{\ell} = -4\hat{i} + \hat{j} - \hat{k}$$

and $\vec{m} = 2\hat{i} - 5\hat{j} - 7\hat{k}$ then the p.v. of a point which lies on both of these lines is

(A) $\hat{i} + 2\hat{j} + \hat{k}$ (B) $2\hat{i} + \hat{j} + \hat{k}$

(C) $\hat{i} + \hat{j} + 2\hat{k}$

(D) non existent s the lines are skew

4. L_1 and L_2 are two lines whose vector equations are

$$L_1 : \vec{r} = \lambda \left[(\cos \theta + \sqrt{3})\hat{i} + (\sqrt{2} \sin \theta)\hat{j} + (\cos \theta - \sqrt{3})\hat{k} \right]$$

$L_2 : \vec{r} = \mu (\hat{a}\hat{i} + \hat{b}\hat{j} + \hat{c}\hat{k})$, Where, λ and μ are scalars and α is the acute angle between L_1 and L_2 . If the angle α is independent of θ then the value of α is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

5. If three lines

$$L_1 : x = y = z \quad L_2 : x = \frac{y}{2} = \frac{z}{3}$$

$$L_3 : \frac{x-1}{a} = \frac{y-1}{b} = \frac{z-1}{c} \text{ form a triangle of}$$

area $\sqrt{6}$ sq.units, then for the point of intersection (a, β, γ) of L_2 and L_3 , $\beta =$

(A) 2 (B) 4 (C) 6 (D) 8

6. Image of the point P with position vector $7\hat{i} - \hat{j} + 2\hat{k}$ in the line whose vector equation is $\vec{r} = 9\hat{i} + 5\hat{j} + 5\hat{k} + \lambda(\hat{i} + 3\hat{j} + 5\hat{k})$ has the position vector.

(A) $(-9, 5, 2)$ (B) $(9, 5, -2)$

(C) $(9, -5, -2)$ (D) none

7. The intercept made by the plane $\vec{r} \cdot \vec{n} = q$ on the x-axis is

(A) $\frac{q}{\hat{i} \cdot \vec{n}}$ (B) $\frac{\hat{i} \cdot \vec{n}}{q}$ (C) $(\hat{i} \cdot \vec{n})q$ (D) $\frac{q}{|\vec{n}|}$

8. ABC is any triangle and O is any point in the plane of the triangle. AO, BO, CO meet the sides BC, CA, AB in D, E, F respectively.

$$\text{Find } \frac{OD}{AD} + \frac{OE}{BE} + \frac{OF}{CF}.$$

(A) 1 (B) 2 (C) -1 (D) -2

9. If from the point P(f, g, h) perpendicular PL, PM be drawn to YZ and ZX planes then the equation of the plane OLM is

$$(A) \frac{x}{f} + \frac{y}{g} - \frac{z}{h} = 0 \quad (B) \frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$$

$$(C) \frac{x}{f} - \frac{y}{g} + \frac{z}{h} = 0 \quad (D) -\frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$$

10. If the distance from point P(1, 1, 1) to the line passing through the points Q(0, 6, 8) and

R (-1, 4, 7) is expressed in the form $\sqrt{\frac{p}{q}}$

where p and q are coprime, then the value

$$\frac{(p+q)(p+q-1)}{2} \text{ equals}$$

(A) 4950 (B) 5050 (C) 5150 (D) none

11. Consider the following 3 lines in space

$$L_1 : \vec{r} = 3\hat{i} - \hat{j} + 2\hat{k} + \lambda(2\hat{i} + 4\hat{j} - \hat{k})$$

$$L_2 : \vec{r} = \hat{i} + \hat{j} - 3\hat{k} + \mu(4\hat{i} + 2\hat{j} + 4\hat{k})$$

$$L_3 : \vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k} + t(2\hat{i} + \hat{j} + 2\hat{k})$$

Then which one of the following pair(s) are in the same plane.

- (A) only L_1L_2 (B) only L_2L_3
(C) only L_3L_1 (D) L_1L_2 and L_2L_3

12. Position vectors of the four angular points of a tetrahedron ABCD are $A(3, -2, 1)$; $B(3, 1, 5)$; $C(4, 0, 3)$ and $D(1, 0, 0)$. Acute angle between the plane faces ADC and ABC is

- (A) $\tan^{-1}\left(\frac{5}{2}\right)$ (B) $\cos^{-1}\left(\frac{2}{5}\right)$
(C) $\operatorname{cosec}^{-1}\left(\frac{5}{2}\right)$ (D) $\cot^{-1}\left(\frac{3}{2}\right)$

13. If a plane passing through the point $(1, 2, 3)$ cuts +ve directions of co-ordinate axes in A, B & C, then the minimum volume of the tetrahedron formed by origin and A, B, C is cubic units

- a) $\frac{9}{2}$ b) 9 c) 18 d) 27

14. A, B, C, D are 4 coplanar points and A', B', C', D' are their projections on any plane. If α is the angle between plane of ABCD and plane of projections then

$$\frac{\text{Volume of tetrahedron } AB'C'D'}{\text{Volume of tetrahedron } A'BCD} =$$

- (A) 1 (B) 2
(C) $2 \cos \alpha$ (D) $\cos \alpha$

15. Let a point R lies on the plane $x - y + z - 3 = 0$ and P be the point $(1, 1, 1)$. A point Q lies on PR such that $PQ^2 + PR^2 = k$ ($\neq 0$) then the equation of locus of Q is

A) $\left[(x-1)^2 + (y-1)^2 + (z-1)^2\right] \left[1 + \frac{4}{(x-y+z-1)^2}\right] = k$

B) $\left[(x-1)^2 + (y-1)^2 + (z-1)^2\right] \left[1 - \frac{4}{(x-y+z-1)^2}\right] = k$

C) $\left[(x-1)^2 + (y-1)^2 + (z-1)^2\right] \left[1 - \frac{4}{(x-y+z-1)^2}\right] = k$

D) $\frac{1}{(x-1)^2} + \frac{1}{(y-1)^2} + \frac{1}{(z-1)^2} + \frac{(x-y+z-1)^2}{4} = k$

16. Let OABC be tetrahedron, Let the mid points of edges OA & OB and OC be A_1, B_1, C_1 respectively while those of edges AB, BC and AC be R, P and Q respectively. If OA is

- (A) $QB_1^2 = RC_1^2$ (B) $QA_1^2 = RC_1^2$
(C) $QC_1^2 = RC_1^2$ (D) None

17. Let $\Delta_1, \Delta_2, \Delta_3$ and Δ_4 be the areas of the triangular faces of tetrahedron and $h_1, h_2, h_3, \& h_4$ be the corresponding altitudes of the tetrahedron, then the minimum value

of $\sum_{1 \leq i < j \leq 4} (\Delta_i h_j) =$ Given volume of the tetrahedron is 5 cubic units.

- (A) 240 (B) 225 (C) 160 (D) 180

18. A line is drawn from the point $P(1, 1, 1)$ and Perpendicular to a line with direction ratios $(1, 1, 1)$ to intersect the plane $x + 2y + 3z = 4$ at Q. The locus of point Q is

A) $\frac{x}{1} = \frac{y-5}{-2} = \frac{z+2}{1}$

$$B) \frac{x}{-2} = \frac{y-5}{1} = \frac{z+2}{1}$$

$$C) x = y = z$$

$$D) \frac{x}{2} = \frac{y}{3} = \frac{z}{5}$$

19. Three positive real numbers x, y, z satisfy the equations

$$x^2 + \sqrt{3}xy + y^2 = 25, y^2 + z^2 = 9$$

and $x^2 + xz + z^2 = 16$. Then the value of

$$xy + 2yz + \sqrt{3}xz \text{ is}$$

$$A) 18$$

$$B) 24$$

$$C) 30$$

$$D) 36$$

20. Three straight lines mutually perpendicular to each other meet in a point P and one of them intersects the x -axis and another intersects the y -axis, while the third line passes through a fixed point $(0, 0, c)$ on the Z -axis. Then the locus of P is

$$A) x^2 + y^2 + z^2 - 2cx = 0$$

$$B) x^2 + y^2 + z^2 - 2cy = 0$$

$$C) x^2 + y^2 + z^2 - 2cz = 0$$

$$D) x^2 + y^2 + z^2 - 2c(x + y + z) = 0$$

21. Perpendiculars are drawn from points on the

line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane

$x + y + z = 3$. The feet of perpendiculars lie on the line

$$A) \frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13} \quad B) \frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$$

$$C) \frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7} \quad D) \frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$$

22. Consider the lines

$$L_1: \frac{x-1}{2} = \frac{y}{1} = \frac{z+3}{1}, L_2: \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$$

and the planes

$$P_1: 7x + y + 2z = 3, P_2: 3x + 5y - 6z = 4. \text{ Let}$$

$ax + by + cz = d$ the equation of the plane

passing through the point of intersection of lines L_1 and L_2 and perpendicular to

planes P_1 and P_2 .

Match List -I with List - II and select the correct answer using the code given below the lists

List - I		List - II	
(P)	$a =$	(1)	13
(Q)	$b =$	(2)	-3
(R)	$c =$	(3)	1
(S)	$d =$	(4)	-2

Codes :

	P	Q	R	S
(A)	3	2	4	1
(B)	1	3	4	2
(C)	3	2	1	4
(D)	2	4	1	3

23. The shortest distance from the point $(1, 2, 3)$

to $x^2 + y^2 + z^2 - xy - yz - zx = 0$ is

$$A) \frac{1}{2} \quad B) 1 \quad C) \sqrt{2} \quad D) \frac{1}{\sqrt{2}}$$

24. A rigid body rotates about an axis through the origin with an angular velocity $10\sqrt{3}$ radians/s if $\vec{\omega}$ points in the direction of $\hat{i} + \hat{j} + \hat{k}$ then the equation to the locus of the points having tangential speed 20 m/sec. is

$$(A) x^2 + y^2 + z^2 - xy - yz - zx - 1 = 0$$

$$(B) x^2 + y^2 + z^2 - 2xy - 2yz - 2zx - 1 = 0$$

$$(C) x^2 + y^2 + z^2 - xy - yz - zx - 2 = 0$$

$$(D) x^2 + y^2 + z^2 - 2xy - 2yz - 2zx - 2 = 0$$

25. A point Q at a distance 3 from the point $P(1, 1, 1)$ lying on the line joining the points $A(0, -1, 3)$ and P has the coordinates

$$(A) (2, 3, -1)$$

$$(B) (4, 7, -5)$$

$$(C) (0, -1, 3)$$

$$(D) (-2, -5, 7)$$

26. Let PM be the perpendicular from the point $P(1, 2, 3)$ to XY plane. If OP makes an angle q with the positive direction of the z -axis and OM makes an angle f with the positive

direction of x-axis, where O is the origin then (q and f are acute angles)

$$(A) \tan \theta = \frac{\sqrt{5}}{3} \quad (B) \sin \theta \sin \phi = \frac{2}{\sqrt{14}}$$

$$(C) \tan \phi = 2 \quad (D) \cos \theta \cos \phi = \frac{1}{\sqrt{14}}$$

27. If the direction ratios of a line are $1+\lambda, 1-\lambda, 2$, and it makes an angle 60° with the y-axis then λ is

$$(A) 1+\sqrt{3} \quad (B) 2+\sqrt{5}$$

$$(C) 1-\sqrt{3} \quad (D) 2-\sqrt{5}$$

28. The line $x+2y-z-3=0, x+3y-z-4=0$ is parallel to

$$(A) XY \text{ plane} \quad (B) YZ \text{ plane}$$

$$(C) ZX \text{ plane} \quad (D) Z\text{-axis}$$

29. A variable plane makes with the coordinate planes, a tetrahedron of constant volume $64k^3$. Then the locus of the centroid of tetrahedron is the surface

$$(A) xyz = 6k^3$$

$$(B) xy + yz + zx = 6k^2$$

$$(C) x^2 + y^2 + z^2 = 8k^2$$

$$(D) x^{-2} + y^{-2} + z^{-2} = 8k^{-2}$$

30. The angle between the line $x+2y+3z=0 = 3x+2y+z$ and the y-axis is

$$(A) \frac{1}{2} \sec^{-1} 3 \quad (B) 2 \sec^{-1} 3$$

$$(C) \cos^{-1} \left(\frac{2}{\sqrt{6}} \right) \quad (D) 2 \sec^{-1} (4)$$

31. If p_1, p_2, p_3 denote the perpendicular distances of the plane $2x-3y+4z+2=0$ from the parallel planes,

$$2x-3y+4z+6=0, 4x-6y+8z+3=0$$

and $2x-3y+4z-6=0$ respectively, then

$$(A) p_1 + 8p_2 - p_3 = 0 \quad (B) p_3 = 16p_2$$

$$(C) 8p_2 = p_1 \quad (D) p_1 + 2p_2 + 3p_3 = \sqrt{29}.$$

32. The line whose vector equations are $\vec{r} = 2\hat{i} - 3\hat{j} + 7\hat{k} + \lambda(2\hat{i} + p\hat{j} + 5\hat{k})$ and

$$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} - p\hat{j} + p\hat{k})$$

are perpendicular for all values of λ and μ if p is equal to

$$(A) -1 \quad (B) 2 \quad (C) 5 \quad (D) 6$$

33. Consider the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z=2$ and

$$\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2} \text{ is}$$

$$(A) \text{Angle between two lines } 90^\circ$$

$$(B) \text{the second line passes through } \left(1, -\frac{3}{2}, -5\right)$$

$$(C) \text{Angle between two lines } 45^\circ$$

$$(D) \text{Angle between two lines is } 30^\circ$$

34. The equation of the bisector planes of an angle between the planes $2x-3y+6z+8=0$ and $x-2y+2z+5=0$

$$(A) x+5y+4z+11=0$$

$$(B) x-5y-4z+11=0$$

$$(C) 13x-23y+32z+59=0$$

$$(D) x+5y+4z+11=0$$

35. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 . Plane P_1 is parallel to the vectors $2\hat{j}+3\hat{k}$ and $4\hat{j}-3\hat{k}$ and that P_2 is parallel to $\hat{j}-\hat{k}$ and $3\hat{i}+3\hat{j}$, then the angle between vector $\vec{A}$ and a given vector $2\hat{i}+\hat{j}-2\hat{k}$ is

$$(A) \frac{\pi}{2} \quad (B) \frac{\pi}{4} \quad (C) \frac{\pi}{6} \quad (D) \frac{3\pi}{4}$$

36. Consider the lines $x=y=z$ and the line $2x+y+z-1=0 = 3x+y+2z-2$ is

$$(A) \text{The shortest distance between the two lines is } \frac{1}{\sqrt{2}}$$

$$(B) \text{The shortest distance between the two lines is } \sqrt{2}$$

$$(C) \text{plane containing 2nd line parallel to 1st line is } y-z+1=0$$

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(D) The shortest distance between the two lines

$$\frac{\sqrt{3}}{2}$$

37. Two systems of rectangular axes have the same origin. If plane cut the intercepts a, b, c on co-ordinate axes for 1st system and intercepts a', b', c' on 2nd system then pick the correct alternatives

(A) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} - \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(B) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2} = 0$

(C) $\frac{1}{a^2} + \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} + \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(D) $\frac{1}{a^2} - \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

38. A line l passing through the origin is perpendicular to the line

$$l_1: (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, -\infty < t < \infty$$

$$l_2: (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, -\infty < s < \infty$$

Then the coordinate (s) of the point (s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 is (are)

A) $\left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right)$ B) $(-1, -1, 0)$

C) $(1, 1, 1)$ D) $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{3}\right)$

39. Consider the planes, $2x + 5y + 3z = 0$
 $x - y + 4z = 2$ and $7y - 5z + 4 = 0$

(A) Planes will meet at a point

(B) Planes will meet on a line

(C) The distance from $(1, 1, 1)$ to one of the

planes to $\frac{\sqrt{2}}{3}$

(D) Planes are equidistant from origin

40. A plane passes through a fixed point (a, b, c) and cuts the axes in A, B, C . The

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locus of a point equidistant from origin, A, B and C must be

(A) $ayz + bzx + cxy = 2xyz$ (B) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$

(C) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$ (D) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 3$

41. Two line $L_1: x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and

$L_2: x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar. Then α can take value (s)

A) 1 B) 2 C) 3 D) 4

COMPREHENSION QUESTIONS

PASSAGE - I

Suppose direction cosines of two lines are given by $ul + vm + wn = 0$ and $a^2l^2 + b^2m^2 + c^2n^2 = 0$, where u, v, w, a, b, c are arbitrary constants and l, m, n are direction cosines of the lines.

On the basis of above information, answer the following questions:

42. For $u = v = w = 1$, both lines satisfies the relation

(A) $(b+c)\left(\frac{n}{l}\right)^2 + 2b\left(\frac{n}{l}\right) + (a+b) = 0$

(B) $(c+a)\left(\frac{l}{m}\right)^2 + 2c\left(\frac{l}{m}\right) + (b+c) = 0$

(C) $(a+b)\left(\frac{m}{n}\right)^2 + 2a\left(\frac{m}{n}\right) + (c+a) = 0$

(D) All of these

43. For $u = v = w = 1$, if $\frac{n_1 n_2}{l_1 l_2} = \left(\frac{a+b}{b+c} \right)$, then

(A) $\frac{m_1 m_2}{l_1 l_2} = \frac{(b+c)}{(c+a)}$ (B) $\frac{m_1 m_2}{l_1 l_2} = \frac{(c+a)}{(b+c)}$

(C) $\frac{m_1 m_2}{l_1 l_2} = \frac{(a+b)}{(c+a)}$ (D) $\frac{m_1 m_2}{l_1 l_2} = \frac{(c+a)}{(a+b)}$

44. For $u = v = w = 1$ and if lines are perpendicular, then

(A) $a + b + c = 0$

(B) $ab + bc + ca = 0$

(C) $ab + bc + ca = 3abc$

(D) $ab + bc + ca = abc$

45. The given lines will be parallel if

(A) $\sum u^2 (b+c) = 0$ (B) $\sum \frac{a^2}{u} = 0$

(C) $\sum \frac{u^2}{a} = 0$ (D) $\sum \frac{(b+c)}{u^2} = 0$

PASSAGE - II

The vector equation of a plane is a relation satisfied by position vectors of all the points on the plane. If P is a plane and $\hat{n}$ is a unit vector through origin which is perpendicular to the plane P then vector equation of the plane must be $\vec{r} \cdot \hat{n} = d$ where d represents perpendicular distance of plane P from origin.

46. If A is a point vector $\vec{a}$ then perpendicular distance of A from the plane $\vec{r} \cdot \hat{n} = d$ must be

(A) $|d + \vec{a} \cdot \hat{n}|$ (B) $|d - \vec{a} \cdot \hat{n}|$

(C) $|\vec{a}| - d$ (D) $|d - \vec{a}|$

47. If $\vec{b}$ be the foot of perpendicular from A to the plane $\vec{r} \cdot \hat{n} = d$ then $\vec{b}$ must be

(A) $\vec{a} + (d - \vec{a} \cdot \hat{n}) \hat{n}$ (B) $\vec{a} - (d - \vec{a} \cdot \hat{n}) \hat{n}$

(C) $\vec{a} + \vec{a} \cdot \hat{n}$ (D) $\vec{a} - \vec{a} \cdot \hat{n}$

48. The position vector of the image of the point $\vec{a}$ in the plane $\vec{r} \cdot \hat{n} = d$ must be ($d \neq 0$)

(A) $-\vec{a} \cdot \hat{n}$ (B) $\vec{a} - 2(d - \vec{a} \cdot \hat{n}) \hat{n}$

(C) $\vec{a} + 2(d - \vec{a} \cdot \hat{n}) \hat{n}$ (D) $\vec{a} + d(-\vec{a} \cdot \hat{n})$

PASSAGE - III

Let the planes $P_1 : 2x - y + z = 2$ and $P_2 : x + 2y - z = 3$ are given. On the basis of the above information, answer the following questions

49. The equation of the plane through the intersection of P_1 and P_2 and the point $(3, 2, 1)$ is

(A) $3x - y + 2z - 9 = 0$

(B) $x - 3y + 2z + 1 = 0$

(C) $2x - 3y + z - 1 = 0$

(D) $4x - 3y + 2z - 8 = 0$

50. Equation of the plane which passes through the point $(-1, 3, 2)$ and is perpendicular to each of the planes P_1 and P_2 is

(A) $x + 3y - 5z + 2 = 0$

(B) $x + 3y + 5z - 18 = 0$

(C) $x - 3y - 5z + 20 = 0$

(D) $x - 3y + 5z = 0$

51. The equation of the acute angle bisector of planes P_1 and P_2 is

(A) $x - 3y + 2z + 1 = 0$

(B) $3x + y - 5 = 0$

(C) $x + 3y - 2z + 1 = 0$

(D) $3x + z + 7 = 0$

52. The equation of the bisector of angle of the planes P_1 and P_2 which is not containing origin, is

(A) $x - 3y + 2z + 1 = 0$

(B) $x + 3y = 5$

(C) $x + 3y + 2z + 2 = 0$

(D) $3x + y = 5$

53. The image of plane P in the plane mirror P_2 is

(A) $x + 7y - 4z + 5 = 0$

(B) $3x + 4y - 5z + 9 = 0$

(C) $7x - y + 4z - 9 = 0$

(D) None of the above

PASSAGE - IV

If α, β, γ are angles made by a line from x, y, z axis respectively, then $\cos \alpha, \cos \beta, \cos \gamma$ are known as direction cosines of a line and represented by l, m, n respectively. Direction ratio are quantities which are

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directly proportional to direction cosines. If

$$\frac{x - \alpha_1}{l_1} = \frac{y - \beta_1}{m_1} = \frac{z - \gamma_1}{n_1} \text{ and}$$

$$\frac{x - \alpha_2}{l_2} = \frac{y - \beta_2}{m_2} = \frac{z - \gamma_2}{n_2} \text{ are two lines, then}$$

angle between them is given by $\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$ and shortest distance between two lines exists along a line which is normal to both of them. $(l_1, m_1, n_1), (l_2, m_2, n_2)$ are direction cosines of the two straight line.

54. Minimum distance between lines

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} \text{ and } \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4} \text{ is}$$

$$(A) \frac{1}{\sqrt{6}} \quad (B) \frac{5}{\sqrt{6}} \quad (C) \frac{7}{\sqrt{6}} \quad (D) \frac{9}{\sqrt{6}}$$

55. If P and Q are points of intersection of line

$$1 - x = \frac{y}{2} = z \text{ with x-axis and plane } x + 2z = 0,$$

then area of triangle OPQ (O is origin) is

$$(A) \frac{\sqrt{3}}{2} \quad (B) 2 \quad (C) \frac{\sqrt{5}}{2} \quad (D) 2\sqrt{2}$$

56. Values of 'a' for which lines whose dc's are connected by the relations $l + am + n = 0$

($a \in \mathbb{R}$) and $2l^2 + m^2 - n^2 = 0$ are parallel, is / are

$$(A) \pm \frac{1}{\sqrt{2}} \quad (B) \pm 2 \quad (C) 2, 3 \quad (D) \pm 1$$

PASSAGE-V

If the three plane $x = y \sin \psi + z \sin \phi$, $y = z \sin \theta + x \sin \psi$, $z = x \sin \phi + y \sin \theta$, intersect in

a line, then where $\theta, \phi \in \left(0, \frac{\pi}{2}\right)$

57. θ, ϕ & ψ satisfy

$$(A) \sin^2 \theta + \sin^2 \phi + \sin^2 \psi = 1$$

$$(B) \sin^2 \theta + \sin^2 \phi + \sin^2 \psi +$$

$$2 \sin \theta \sin \phi \sin \psi = 1$$

$$(C) \cos^2 \theta + \cos^2 \phi + \cos^2 \psi = 1$$

$$(D) \sin^2 \theta + \sin^2 \phi + \sin^2 \psi = 1$$

$$58. \theta + \phi + \psi =$$

$$(A) 90^\circ \quad (B) 120^\circ \quad (C) 150^\circ \quad (D) 180^\circ$$

59. Equation of their common line is

$$(A) \frac{x}{\sin \theta} = \frac{y}{\sin \phi} = \frac{z}{\sin \psi}$$

$$(B) \frac{x}{\cos \theta} = \frac{y}{\cos \phi} = \frac{z}{\cos \psi}$$

$$(C) \frac{x}{\tan \theta} = \frac{y}{\tan \phi} = \frac{z}{\tan \psi}$$

$$(D) \text{None}$$

PASSAGE - VI

A variable point P (α, β, γ) moves on a fixed

plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. Then plane through P and perpendicular to OP meets the coordinate axes in A, B, C. If the planes through A, B, C respectively parallel to co-ordinate planes YOZ, XOZ, meet in point Q then

60. A =

$$(A) (\alpha, 0, 0) \quad (B) \left(\frac{d}{\alpha + \beta + \gamma}, 0, 0 \right)$$

$$(C) \left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, 0, 0 \right) \quad (D) \text{None}$$

61. If the surface generated by Q passes

through (1,1,1), then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} =$

$$(A) 1$$

$$(B) 2$$

$$(C) 3$$

$$(D) \text{None of these}$$

PASSAGE-VII

From any point P (a, b, c) perpendiculars PM & PN drawn to zx and xy-plane respectively. Let α, β, γ be the angles which OP makes with coordinate planes and θ be the angle which OP makes with the plane OMN must be

62. Equation of plane OMN must be

$$(A) \frac{x}{a} - \frac{y}{b} + \frac{z}{c} = 0 \quad (B) \frac{x}{a} + \frac{y}{b} - \frac{z}{c} = 0$$

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(C) $\frac{x}{a} - \frac{y}{b} + \frac{z}{c} = 0$ (D) None

63. $\sin \theta$ must be equal to

(A) $\frac{abc}{\sqrt{a^2 + b^2 c^2} \sqrt{a^2 b^2 + b^2 c^2 + c^2 a^2}}$

(B) $\frac{ab + bc + ca}{a^2 + b^2 + c^2}$

(C) $\frac{a + b + c}{a^2 + b^2 + c^2}$

(D) None

64. $\operatorname{Cosec}^2 \theta =$

(A) $\cot^2 \alpha + \cot^2 \beta + \cot^2 \gamma$

(B) $\cos ec^2 \alpha + \cos ec^2 \beta + \cos ec^2 \gamma$

(C) $\cos ec \alpha + \cos ec \beta + \cos ec \gamma$

(D) None

PARAGRAPH - VIII

Consider the lines represented parametrically as

$L_1: x = 1 - 2t, y = t, z = -1 + t$

$L_2: x = 4 + S, y = 5 + 4S, z = -2 - S$

Let π be the plane containing the line L_2 and parallel to line L_1

65. The acute angle between the lines L_1 and L_2 is

A) $\cos^{-1} \left(\frac{1}{18} \right)$ B) $\cos^{-1} \left(\frac{1}{6\sqrt{3}} \right)$

C) $\cos^{-1} \left(\frac{1}{3\sqrt{2}} \right)$ D) $\cos^{-1} \left(\frac{1}{3\sqrt{6}} \right)$

66. The equation of plane π is

A) $5x + y + 9z - 7 = 0$

B) $9x - 5y - z - 13 = 0$

C) $2x - 3y - 4z - 15 = 0$

D) $5x - y + 9z + 3 = 0$

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67. The distance between the plane π and the line L_1 is

A) $\frac{17}{\sqrt{19}}$ B) $\frac{3}{\sqrt{87}}$ C) $\frac{1}{\sqrt{107}}$ D) $\frac{11}{\sqrt{107}}$

PARAGRAPH - IX

Let A denote the plane consisting of all points that are equidistant from the points $P(-4, 2, 1)$ and $Q(2, -4, 3)$ and B be the plane, $x - y + cz = 1$ where $c \in \mathbb{R}$

68. If the angle between the planes A and B is 45° then the product of all possible values of c is

A) -17 B) -2 C) 17 D) $\frac{24}{17}$

69. If the line L with equation $\frac{x-2}{1} = \frac{y-1}{3} = \frac{z-5}{-1}$ intersects the plane A at the point $M(\lambda, \mu, \nu)$, then coordinate of M is

A) $\left(\frac{8}{7}, \frac{11}{7}, \frac{41}{7} \right)$ B) $\left(-\frac{8}{7}, -\frac{11}{7}, \frac{41}{7} \right)$

C) $\left(\frac{8}{7}, -\frac{11}{7}, \frac{41}{7} \right)$ D) None of these

PARAGRAPH - X

The line of greatest slope on an inclined plane P_1 is the line in the plane P_1 which is perpendicular to the line of intersection of the plane P_1 and a horizontal plane P_2 .

70. Assume the plane $4x - 3y + 7z = 0$ to be horizontal, the direction cosine of the line of greatest slope in the plane $2x + y - 5z = 0$ are

A) $\frac{3}{\sqrt{11}}, \frac{-1}{\sqrt{11}}, \frac{1}{\sqrt{11}}$ B) $\frac{3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{-1}{\sqrt{11}}$

C) $\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}$ D) None of these

71. The coordinates of a point on the plane $2x + y - 5z = 0$, $2\sqrt{11}$ units away from the line of intersection of

$2x + y - 5z = 0$ and $4x - 3y + 7z = 0$ are

A) $(6, 2, -2)$ B) $(3, 1, -1)$

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- C) $(6, -2, 2)$ D) $(1, 3, -1)$

PARAGRAPH - XI

In three dimensions there may be more than one point, which are equidistant from three given non-collinear points A, B, C. One of these points will be circumcentre of the triangle ABC

72. The circumcentre of the triangle ABC where A, B, C are $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ will lie in the plane

A) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ B) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 2$

C) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$ D) none of these

73. y coordinate of the circumcentre of triangle ABC must be

A) $\frac{ac}{a+b+c}$

B) $\frac{a^2c^2 - b^4}{a^3 + b^3 + c^3}$

C) $\left(\frac{b^2c^2 + a^2b^2 - a^2c^2}{a^2b^2 + b^2c^2 + c^2a^2} \right)$

D) $\frac{b^3(a^2 + c^2)}{2(b^2c^2 + a^2c^2 + a^2b^2)}$

74. The y coordinate of orthocentre of the triangle ABC

A) $\left(\frac{3a^2c^2 - a^2b^2 - b^2c^2}{a^2b^2 + b^2c^2 + c^2a^2} \right)$

B) $\frac{ab + b^2 - ac}{a + b + c}$

C) $b - \frac{2(a^2c^2 - b)}{a^3 + b^3 + c^3}$

D) $\frac{a^2bc^2}{b^2c^2 + c^2a^2 + a^2b^2}$

ASSERTION & REASON QUESTIONS

- (A) Statement -1 is true, statement -2 is true, statement -2 is a correct explanation for statement-1
(B) Statement-1 is true, statement-2 is true, statement -2 is not correct explanation for

statement-1

- (C) Statement-1 is true, statement-2 is false
(D) Statement -1 is false, statement-2 is true

75. Statement - 1: If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors, then the length of projection of vector $\vec{a}$ in the plane of vectors

$\vec{b}$ and $\vec{c}$ may be $\frac{|\vec{a} \times (\vec{b} \times \vec{c})|}{|\vec{b} \times \vec{c}|}$

Statement - 2: $\hat{n}$ = unit vector normal to plane

$\vec{b}$ and $\vec{c}$ is $\frac{\vec{b} \times \vec{c}}{|\vec{b} \times \vec{c}|}$ & projection of $\vec{a}$ in the

plane of $\vec{b}$ and $\vec{c}$ is $\frac{|\vec{a} \times (\vec{b} \times \vec{c})|}{|\vec{b} \times \vec{c}|}$

76. Statement-1: If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then equation $\vec{r} \times (2\hat{i} - \hat{j} + 3\hat{k}) = 3\hat{i} + \hat{k}$ represents a straight line

Statement-2: If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then equation $\vec{r} \times (\hat{i} + 2\hat{j} - 3\hat{k}) = 2\hat{i} - \hat{j}$ represents a straight line

77. Statement 1 : Planes parallel to x-axis and passing through the point $(2, 1, 3)$ will not be at a fixed distance from the x-axis.

because

Statement 2 : Such planes will be tangential to a cylinder with its axis as x-axis.

78. Statement 1 : The equation $2x^2 - 6y^2 + 4z^2 + 18yz + 2zx + xy = 0$ represents a pair of perpendicular planes.

Statement 2 : A pair of planes given by $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ are perpendicular, if $a + b + c = 0$

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79. Statement 1 : The shortest distance between

the skew lines $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$ and

$\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$ is 9.

Statement 2 : Two lines are skew lines if there exists no plane passing through them.

80. Statement 1 : The equation of the plane through the intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$ and the point $(4, 4, 4)$ is $29x + 23y + 17z = 276$.

Statement 2 : Equation of the plane through the line of intersection of the planes

$P_1 = 0$ and $P_2 = 0$ is $P_1 + \lambda P_2 = 0$

81. Consider the planes

$P_1 : x - y + z = 1$; $P_2 : x + y - z = -1$ and

$P_3 : x - 3y + 3z = 2$

Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3 , P_3 and P_1 , P_1 and P_2 respectively.

Statement 1 : At least two of the lines L_1, L_2 and L_3 are non-parallel.

Statement II : The three planes do not have a common point. [IIT-2008]

MATRIX-MATCHING QUESTIONS

This section contains 1 questions. Each questions contain statements given in two columns, which have to be matched. The statements in Column I are labeled A, B, C and D while the statements in Column II are labelled p, q, r, s and t. Any given statement

in Column I can have correct matching with ONE OR MORE statement(s) in Column II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example. If the correct matches are A-p, s and t, B-q and r, C-p and q, and D-s and t, then the correct darkening of bubbles will look like the following :

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

82. Match the statements/expressions given in Column I with the values given in Column II

(A) A line is perpendicular to $x + 2y + 2z = 0$ and passes through $(0, 1, 0)$. The perpendicular distance of this line from $(0, 0, 0)$ is

(B) A plane passes through $(1, -2, 1)$ and is perpendicular to two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$. The distance of the plane from the point $(1, 2, 2)$ is

(C) The volume of the tetrahedron whose vertices are $A(3, 7, 4)$, $B(5, -2, 3)$, $C(-4, 5, 6)$ and $D(1, 2, 3)$ is

(D) The perpendicular distance between

$2x + 2y - z + 1 = 0$ and $x + y - \frac{z}{2} + 2 = 0$ is

Column II

(p) 2.82

(q) 1.00

(r) 0.74

(s) 15.33

83. Match the statements/expressions given in Column I with the values given in Column II

(A) Let $\vec{a} = 2\hat{i} + 3\hat{j} - 5\hat{k}$, $\vec{b} = 3\hat{i} - 6\hat{j} + 7\hat{k}$, $\vec{c} = 12\hat{i} + 5\hat{j}$. A vector in the plane of $\vec{a}$ and $\vec{b}$

whose projection on $\vec{c}$ is $\frac{1}{13}$ is

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(B) Let $\vec{a}$ be a vector parallel to the line of intersection of the planes π_1 and π_2 , through the origin. π_1 is parallel to the vectors $3\hat{i} + 2\hat{j}$ and $3\hat{j} - 4\hat{k}$. π_2 is parallel

to $-\hat{j} + \hat{k}$ and $5\hat{i} + 5\hat{j}$. Then $\vec{a}$ is

(C) Point (a, b, c) lies on the plane, $x + y + z =$

2. Let $\vec{p} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\hat{j} \times (\hat{j} \times \vec{p}) = \vec{0}$ and

$\hat{k} \times (\hat{k} \times \vec{p}) = \vec{0}$ Then $\vec{p}$ is

(D) A unit vector coplanar with and perpendicular to $\hat{i} + \hat{j} + \hat{k}$ is

Column II

(p) a null vector

(q) $-\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}$

(r) $-\frac{1}{3}(51\hat{i} - 123\hat{j} + 148\hat{k})$

(s) $15\lambda\hat{i} - 5\lambda\hat{j} + 20\lambda\hat{k}$

84. Match the statements/expressions given in Column I with the values given in Column II

(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$

$L_2 : \vec{r} = (2, 1, -3) + \lambda(2, 2, -10)$

(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$

$L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$

(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$

$L_2 : x = 1 + 2s, y = 4 - 3s, z = s$

(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

$L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$

Column II

(p) non coplanar lines

(q) lines lie in a unique plane

(r) infinite planes containing both the lines

(s) lines are not intersecting.

85. P(0, 3, -2); Q(3, 7, -1) and R(1, -3, 1) are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i} + \hat{k}$

Column-I

(A) Perpendicular distance of P from L_2

(B) Shortest distance between L_1 and L_2

(C) Area of the triangle PQR

(D) Distance from (0, 0, 0) to the plane PQR

Column-II

(p) $7\sqrt{3}$

(q) 2

(r) 6

(s) $\frac{19}{\sqrt{147}}$

86. Consider the following linear equations

$ax + by + cz = 0, \quad bx + cy + az = 0,$

$cx + ay + bz = 0 \quad [\text{IIT-JEE 2007}]$

Column-I

(A) $a + b + c \neq 0$ and

$a^2 + b^2 + c^2 = ab + bc + ca$

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

Column-II

(p) the equations represent planes meeting only at a single point

(q) the equations represent the line $x = y = z$

(r) the equations represent identical planes

(s) the equations represent the whole of the three dimensional space.

INTEGER QUESTIONS

The answer to each of the question is a single digit integer, ranging from 0 to 9. The

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appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X,Y,Z and W (say) are 6,0,9 and 2, respectively, then the correct darkening of bubbles will look like the following:

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

87. Find the shortest distance of plane parallel to z-axis and containing line

$$x + y + 2z - 3 = 0 = 2x + 3y + 4z - 4 \text{ from } z\text{-axis.}$$

88. The shortest distance between origin and a point on the space curve $x = 2 \sin t, y = 2 \cos t, z = 3t$ is.

89. Value of λ such that the planes $x - y + z + 1 = 0$, $\lambda x + 3y + 2z - 3 = 0$, $3x + \lambda y + z - 2 = 0$ form a triangular prism must be.

90. The equation of the plane containing the line $2x - y + z - 3 = 0$, $3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point $(2,1,-1)$ is $62x + 29y + 19z + k = 0$. Then the sum of digits of K is

[IIT-JEE 2005]

91. A plane is parallel to two lines whose direction ratios are $(1,0,-1)$ and $(-1,1,0)$ and it contains the point $(1,1,1)$. If it cuts coordinate axes at A, B, C. If V is the

volume of the tetrahedron OABC then $2V$ is

[IIT-JEE 2004]

92. If the centroid of the tetrahedron OABC where A,B,C are the points $(a,2,3)$, $(1,b,2)$ and $(2,1,c)$ be $(1,2,3)$ and the point (a,b,c) is at distance $5\sqrt{\lambda}$ from origin, then λ^2 must be equal to.

93. If the triangle ABC whose vertices are $A(-1,1,1)$, $B(1,-1,1)$ and $C(1,1,-1)$ is projected on XY plane and the area of the projected triangle is m then find $\frac{m(m+1)}{2}$.

94. If the circumcentre of the triangle whose vertices are $(3,2,-5)$, $(-3,8,-5)$ and $(-3,2,1)$ is $(-1,\lambda,-3)$ the integer λ must be equal to.

95. The plane $2x - 2y + z = 3$ is rotated about the line where it cuts the xy plane by an acute angle α . If the new position of the plane contains the point $(3,1,1)$ then the value of $9 \cos \alpha$ is

KEY LEVEL -VI

SINGLE ANSWER QUESTIONS

- | | | | |
|--------------|------------------------|--------------|---------|
| 1. (B) | 2. (C) | 3. (A) | 4. (A) |
| 5. (B) | 6. (B) | 7. (A) | 8. (A) |
| 9. (A) | 10. (A) | 11. (D) | 12. (A) |
| 13. (D) | 14. (A) | 15. (A) | 16. (A) |
| 17. (D) | 18. (A) | 19. (B) | 20. (C) |
| 21. (D) | 22. (A) | 23. (C) | 24. (C) |
| 25. (A) | 26. (A), (B), (C) | | |
| 27. (A), (D) | 28. (C) | 29. (A) | |
| 30. (A), (C) | 31. (A), (B), (C), (D) | | |
| 32. (A), (D) | 33. (A), (C) | | |
| 34. (B), (D) | 35. (B), (D) | | |
| 36. (A), (C) | 37. (A) | 38. (B), (D) | |
| 39. (A), (C) | 40. (A), (C) | | |
| 41. (A), (D) | 42. (D) | | |
| 43. (B) | 44. (A) | 45. (C) | |
| 46. (B) | 47. (A) | 48. (C) | 49. (B) |
| 50. (C) | 51. (A) | 52. (D) | 53. (C) |
| 54. (A) | 55. (C) | 56. (A) | 57. (A) |
| 58. (A) | 59. (B) | 60. (B) | 61. (B) |
| 62. (A) | 63. (B) | 64. (C) | 65. (B) |
| 66. (A) | 67. (D) | 68. (B) | 69. (C) |
| 70. (A) | 71. (C) | 72. (A) | 73. (D) |
| 74. (D) | 75. (C) | 76. (B) | 77. (A) |
| 78. (B) | 79. (B) | 80. (A) | 81. (D) |

MATRIX MATCH QUESTIONS

82. (A) -(r); (B) -(p); (C) -(s); (D) -(q)
 83. (A) -(r); (B) -(s); (C) -(p); (D) -(q)
 84. (A) -(r); (B) -(q); (C) -(q,s); (D) -(p,s)
 85. (A) -(r); (B) -(q); (C) -(p); (D) -(s)
 86. (A) -(r); (B) -(q); (C) -(p); (D) -(s)

INTEGER QUESTIONS

87. 2 88. 2 89. 4 90. 9
 91. 9 92. 9 93. 3 94. 4
 95. 7

HINTS

LEVEL -VI

SINGLE ANSWER QUESTIONS

1. $\overrightarrow{AB} = \lambda \overrightarrow{BC}$

$$\left[(a-1)\hat{i} + (2-a)\hat{j} + (b-b)\hat{k} \right] = \lambda \left[(a-a)\hat{i} + (b-2)\hat{j} + (b-3)\hat{k} \right]$$

$$\Rightarrow a-1=0; 2-a=\lambda(b-2); \lambda(b-3)=0$$

$$\Rightarrow a=1; b=3; \lambda=1$$

$$\Rightarrow a+b=4$$
2. We can prove that collinearity of the points P, Q, R by showing that $PQ + PR = QR$, so that the point P lies on (within) the segment of the line QR. Alternatively, we may proceed as follows. Supposing that the points P, Q, R are collinear, let the point Q divide the line segment PR in the ratio $m_1 : m_2$. Then the coordinates of Q are
- $$\left(\frac{4m_1 + 2m_2}{m_1 + m_2}, \frac{7m_1 + 5m_2}{m_1 + m_2}, \frac{-6m_1 - 4m_2}{m_1 + m_2} \right)$$
- We can find the ratio $m_1 : m_2$ by equating any one of these coordinates to the given coordinates of Q. Thus equating the x-coordinates, we get $4m_1 + 2m_2 = m_1 + m_2$, whence $m_1 : m_2 = -1 : 3$
3. $\vec{r} = 5\hat{i} + \hat{j} + 2\hat{k} + \lambda(-4\hat{i} + \hat{j} - \hat{k})$
 $\vec{r} = (-\hat{i} + 7\hat{j} + 8\hat{k}) + \mu(2\hat{i} - 5\hat{j} - 7\hat{k})$
 If we place $\vec{r} = \hat{i} + 2\hat{j} + \hat{k}$
 it is satisfying both
4. L_1 and L_2 are two lines
- $$\cos \alpha = \frac{a(\cos \theta + \sqrt{3}) + b(\sqrt{2} \sin \theta) + c(\cos \theta - \sqrt{3})}{\sqrt{a^2 + b^2 + c^2} \sqrt{(\cos \theta + \sqrt{3})^2 + (\sqrt{2} \sin \theta)^2 + (\cos \theta - \sqrt{3})^2}}$$
- It is independent of θ .
 $a + c = 0 \Rightarrow a = -c$
 $b = 0$
- $$\cos \alpha = \frac{[a - (-a)]\sqrt{3}}{\sqrt{a^2 + a^2 + a^2}\sqrt{8}} = \frac{\sqrt{3}}{2} \Rightarrow \alpha = \frac{\pi}{6}$$
5. Let

$$L_2: x = \frac{y}{z} = \frac{z}{3} = \lambda \text{ (say)} \Rightarrow (\alpha, \beta, \gamma) = (\lambda, 2\lambda, 3\lambda)$$

$$\Delta = \sqrt{6} \Rightarrow \lambda = 2(\alpha, \beta, \gamma) = (2, 4, 6)$$

6. $\left[\frac{x_1+7}{2}, \frac{y_1-1}{2}, \frac{z_1+2}{2} \right]$ is lying on line

$$\frac{\frac{x_1+7}{2}-9}{1} = \frac{\frac{y_1-1}{2}-5}{3} = \frac{\frac{z_1+2}{2}-5}{5}$$

$$\Rightarrow \frac{x_1-11}{1} = \frac{y_1-11}{3} = \frac{z_1-8}{5} = \lambda$$

...(i)

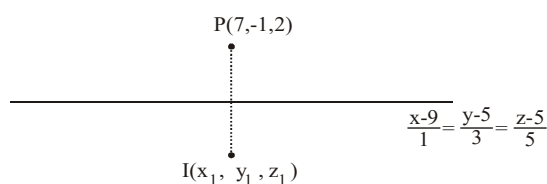
$$\Rightarrow (x_1 = \lambda + 11, y_1 = 3\lambda + 11, z_1 = 5\lambda + 8)$$

as PI is perpendicular to given line.

$$(x_1 - 7) \cdot 1 + (y_1 + 1) \cdot 3 + (z_1 - 2) \cdot 5 = 0$$

$$\Rightarrow (\lambda + 4) \cdot 1 + (3\lambda + 12) \cdot 3 + (5\lambda + 6) \cdot 5 = 0$$

$$35\lambda + 70 = 0 \Rightarrow \lambda = -2 \Rightarrow I(9, 5, -2)$$



7. $(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (n_1\hat{i} + n_2\hat{j} + n_3\hat{k}) = q$

$$n_1x + n_2y + n_3z = q$$

for X intercept put $y = z = 0$

$$x = \frac{q}{n_1} = \frac{q}{\hat{i} \cdot \vec{n}}$$

8. Take O as the origin.

Let $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of the vertices A, B, C of the triangle.

Since $\vec{a}, \vec{b}, \vec{c}$ are coplanar there must exist a relation of the form

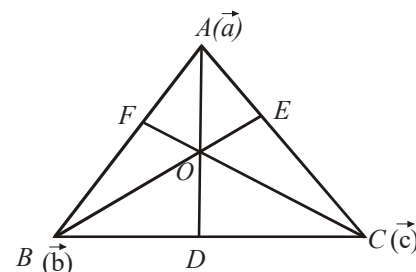
$$x\vec{a} + y\vec{b} + z\vec{c} = \vec{0},$$

$$(x, y, z \in \mathbb{R}, \text{ not all zero}) \quad \dots (i)$$

$$\therefore y\vec{b} + z\vec{c} = -x\vec{a}$$

$$\Rightarrow \frac{y\vec{b} + z\vec{c}}{y+z} = -\frac{x}{y+z}\vec{a}$$

Now $\frac{y\vec{b} + z\vec{c}}{y+z}$ is a point on the line BC.



Equation of $\vec{OA}$ is $\vec{r} = -t\vec{a}$

Thus $-\frac{x}{y+a}\vec{a}$ is a point of AD.

From (ii), $\vec{OD} = \frac{-x}{y+z}\vec{a}$

$$\therefore \frac{OD}{AD} = \frac{|\vec{OD}|}{|\vec{AD}|} = \left(-\frac{x}{y+z} \right) \times \left(-\frac{y+z}{x+y+z} \right) = \frac{x}{x+y+z}$$

Similarly, $\frac{OE}{BE} = \frac{y}{x+y+z}$ and $\frac{OF}{CF} = \frac{z}{x+y+z}$

Adding, $\frac{OD}{AD} + \frac{OE}{BE} + \frac{OF}{CF} = \frac{x+y+z}{x+y+z} = 1$

9. O(0,0,0); L(0,g,h); M(f,0,h)

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ f & 0 & h \\ 0 & g & h \end{vmatrix} = -(gh)\hat{i} - (fh)\hat{j} + (fg)\hat{k}$$

Equation of plane

$$x(-gh) + y(-fh) + z(fg) = 0 \Rightarrow \frac{x}{f} + \frac{y}{g} - \frac{z}{h} = 0$$

10 Equation of RQ is $\frac{x-0}{0-(-1)} = \frac{y-6}{6-4} = \frac{z-8}{8-7}$

$$\Rightarrow \frac{x}{1} = \frac{y-6}{2} = \frac{z-8}{1} = \lambda$$

$$1(\lambda-1) + 2(2\lambda+6-1) + 1(\lambda+8-1) = 0$$

$$6\lambda + 16 = 0; \quad \lambda = -\frac{8}{3}$$

$$M\left(-\frac{8}{3}, \frac{2}{3}, \frac{16}{3}\right)$$

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$$PM = \sqrt{\left(\frac{11}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{13}{3}\right)^2} = \sqrt{\frac{291}{9}} = \sqrt{\frac{97}{3}}$$

$$\frac{(p+q)(p+q-1)}{2} = \frac{(97+3)(97+3-1)}{2} = 4950$$

11. Consider L_1 & L_2

$$\begin{vmatrix} 3-1 & -1-1 & 2+3 \\ 2 & 4 & -1 \\ 4 & 2 & 4 \end{vmatrix} = 2(18) + 2(12) + 5(4-16) = 0$$

Lines are intersecting hence coplanar L_2 & L_3 are parallel hence coplanar.

12. Plane ADC

$$\vec{n}_1 = \vec{AD} \times \vec{AC}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & -1 \\ 1 & 2 & 2 \end{vmatrix} = 6\hat{i} + 3\hat{j} - 6\hat{k} = 3(2\hat{i} + \hat{j} - 2\hat{k})$$

Plane ABC

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 3 & 4 \\ 1 & 2 & 2 \end{vmatrix} = -2\hat{i} + 4\hat{j} - 3\hat{k}$$

$$\cos \theta = \frac{2(-2) + 1(4) + (-2)(-3)}{\sqrt{2^2 + 1^2 + 2^2} \sqrt{2^2 + 4^2 + 3^2}} = \frac{2}{\sqrt{29}}$$

$$\tan \theta = \frac{5}{2} \Rightarrow \theta = \tan^{-1}\left(\frac{5}{2}\right)$$

13. Let $A(\alpha, 0, 0)$, $B(0, b, 0)$ & $C(0, 0, c)$ $a, b, c > 0$

$$\frac{1}{a} + \frac{2}{b} + \frac{3}{c} = 1$$

$$\frac{abc}{6} \geq \left(\frac{1}{\frac{1}{a} + \frac{2}{b} + \frac{3}{c}} \right)^3$$

14. Let $A(x_1, y_1, z_1)$

$$B(x_2, y_2, z_2), C(x_3, y_3, z_4), D(x_4, y_4, z_4)$$

JEE ADVANCED - VOL - IV

Let the plane of projection be xy-plane then $A'(x_1, y_1, 0)$ $B'(x_2, y_2, 0)$ $C'(x_3, y_3, 0)$ and $D'(x_4, y_4, 0)$

$$V_1 = \text{volume of } AB'C'D' = \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x & y_2 & 0 & 1 \\ x_3 & y_3 & 0 & 1 \\ x_4 & y_4 & 0 & 1 \end{vmatrix}$$

and $V_2 = \text{volume } A'BCD$

$$\frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x & y_2 & 0 & 1 \\ x_3 & y_3 & 0 & 1 \\ x_4 & y_4 & 0 & 1 \end{vmatrix}$$

$$V_1 = \text{volume of } AB'C'D' = \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x & y_2 & 0 & 1 \\ x_3 & y_3 & 0 & 1 \\ x_4 & y_4 & 0 & 1 \end{vmatrix}$$

and $V_2 = \text{volume } A'BCD$

$$\frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x & y_2 & 0 & 1 \\ x_3 & y_3 & 0 & 1 \\ x_4 & y_4 & 0 & 1 \end{vmatrix}$$

- 15) Let Q be (α, β, γ) , then

$$PQ^2 = (\alpha-1)^2 + (\beta-1)^2 + (\gamma-1)^2 = r_2^2 \text{ where}$$

$PQ = r_2$ Then $PR = r_1$ l, m, n be the direction cosines of the line PR , then R is $(1 + lr_1, 1 + mr_1, 1 + nr_1)$

$$R \text{ lies on the plane so } r_1 = \frac{2}{l-m+n}$$

$$\text{Also, } Q \text{ is } (1 + lr_2, 1 + mr_2, 1 + nr_2)$$

$$\Rightarrow \frac{\alpha-1}{r_2} = l, \frac{\beta-1}{r_2} = m, \frac{\gamma-1}{r_2} = n$$

$$\Rightarrow r_1 = \frac{2r_2}{\alpha - \beta + \gamma - 1}$$

$$\text{Now, } r_1^2 + r_2^2 = k$$

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$$\Rightarrow r_2^2 \left[1 + \frac{4}{(\alpha + \beta + \gamma - 1)^2} \right] = k$$

Locus of Q is

$$\left[(x-1)^2 + (y-1)^2 + (z-1)^2 \right] \left[1 + \frac{4}{(x-y+z-1)^2} \right] = k$$

16. $OA \perp BC$

$$\Rightarrow x_1(x_2 - x_3) + y_1(y_2 - y_3) + z_1(z_2 - z_3) = 0$$

$$B_1 = \left(\frac{x_2}{2}, \frac{y_2}{2}, \frac{z_2}{2} \right), Q = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

$$(QB_1)^2 = \frac{1}{4} \left[(x_1 + x_2 - x_2)^2 + (y_1 + y_2 - y_2)^2 + (z_1 + z_2 - z_2)^2 \right]$$

$$RC_1^2 = \frac{1}{4} \left[(x_1 + x_2 - x_3)^2 + (y_1 + y_2 - y_3)^2 + (z_1 + z_2 - z_3)^2 \right]$$

$$\text{If } QB_1^2 = (RC_1)^2$$

$$\Rightarrow \sum x_1(x_2 - x_3) = 0 \text{ Which is true}$$

17. Conceptual

18. Locus of Q is the line of intersection of the plane $x + 2y + 3z = 4$ and

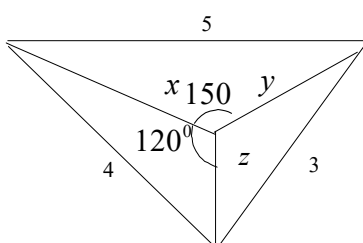
$$1(x-1) + 1(y-1) + 1(z-1) = 0$$

$$\text{then line is } \frac{x}{1} = \frac{y-5}{-2} = \frac{z+2}{1}$$

19. Given

$$x^2 + y^2 - 2xy \cos \frac{5\pi}{6} = 25;$$

$$x^2 + z^2 - 2xz \cos \frac{2\pi}{3} = 16 \text{ and } y^2 + z^2 = 9$$



20. Let $p = (x_1, y_1, z_1)$ and Let $A(a, 0, 0), B(0, b, 0)$ be point of intersection of the lines through P. Given $C = (0, 0, c)$ be the fixed point through which the 3rd line passes.

Now $PA \perp PB, PA \perp PC, PB \perp PC$

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$$x_1(x_1 - a) + y_1(y_1 - b) + z_1^2 = 0 \text{ --- (1)}$$

$$x_1(x_1 - a) + y_1^2 + z_1(z_1 - c) = 0 \text{ --- (2)}$$

$$x_1^2 + y_1(y_1 - b) + z_1(z_1 - c) = 0 \text{ --- (3)}$$

from (2), (3), $ax_1 = by_1$

$$\text{from (1) } ax_1 = \frac{x_1^2 + y_1^2 + z_1^2}{2}$$

Substituting in (2) we get

$$x_1^2 + y_1^2 + z_1^2 - 2cz_1 = 0$$

21. Let $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3} = k$ then

$(2k-2, -k-1, -13k)$ is a point on the line

Let the foot of the perpendicular from this point on the plane $x + y + z - 3 = 0$ be

(α, β, γ) then it is given by

$$\frac{\alpha - (2k-2)}{1} = \frac{\beta - (-k-1)}{1} = \frac{\gamma - (-13k)}{1} = \frac{(2k-2) + (-k-1) + (-13k) - 3}{3} = \frac{6-4k}{3}$$

Thus, $\alpha = \frac{2k}{3}, \beta = 1 - \frac{7k}{3}, \gamma = 2 + \frac{5k}{3}$ then the line is given by

$$\frac{x}{2/3} = \frac{y-1}{-7/3} = \frac{z-2}{5/3} \text{ i.e., } \frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}.$$

22) The normal to the plane is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix} = -16\hat{i} - 48\hat{j} + 32\hat{k} \therefore \vec{h} = \hat{i} - 3\hat{j} - 2\hat{k}$$

The point of intersection of L_1 and L_2 is given by

$$2\lambda + 1 = \mu + 4$$

$$-\lambda = \mu - 3$$

$$1 = 3\mu - 2$$

So, $\mu = 1$

$\therefore$ point of intersection is $(5, -2, -1)$

The equation of desired plane is

$$(x-5) - 3(y+2) - 2(z+1) = 0 \Rightarrow x - 3y - 2z = 13$$

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23. $x^2 + y^2 + z^2 - xy - yz - zx = 0$

let $x = y = z = r$

$1(r-1) + 1(r-2) + 1(r-3) = 0$

$3r - 6 = 0$

$r = 2$

The point on the line is (2,2,2)

Distance = $\sqrt{1+0^2+1^2} = \sqrt{2}$

24. $\vec{\omega} = \frac{10\sqrt{3}(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$

$\vec{v} = \vec{r} \times \vec{\omega}$

$= 10 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 10[(y-z)\hat{i} - (x-z)\hat{j} + (x-y)\hat{k}]$

$10^2[(y-z)^2 + (x-z)^2 + (x-y)^2] = 20^2$

$\Rightarrow x^2 + y^2 + z^2 - xy - yz - 2 = 0$

MULTI ANSWER QUESTIONS

25. If Q is (α, β, γ) then (α, β, γ) , (1, 1, 1) and (0, -1, 3) are collinear.

$\therefore \frac{\alpha-1}{1-0} = \frac{\beta-1}{1-(-1)} = \frac{\gamma-1}{1-3}$. Also PQ = 3

$\Rightarrow (\alpha-1)^2 + (\beta-1)^2 + (\gamma-1)^2 = 9$

Solve these equations for α, β, γ .

26. Here, P be (x, y, z)

Then, $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$,

$z = r \cos \theta$

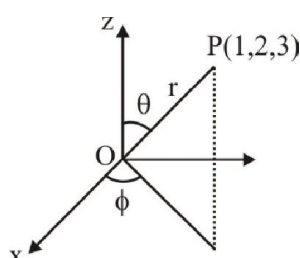
$\Rightarrow 1 = r \sin \theta \cos \phi, 2 = r \sin \theta \sin \phi, 3 = r \cos \theta$

...(i)

$\Rightarrow 1^2 + 2^2 + 3^2 = r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi + r^2 \cos^2 \theta$

$= r^2 \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \cos^2 \theta = r^2$

$\Rightarrow r = \pm \sqrt{14}$



From Eq. (i), we have

$\sin \theta \cos \phi = \pm \frac{1}{\sqrt{14}}$

$\sin \theta \sin \phi = \frac{2}{\sqrt{14}}, \cos \theta = \frac{3}{\sqrt{14}}$

(neglecting -ve sign assuming acute angles)

$\therefore \frac{\sin \theta \sin \phi}{\sin \theta \cos \phi} = \frac{2}{1}$

and $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{5}}{3}$

$\Rightarrow \tan \phi = 2$ and $\tan \theta = \frac{\sqrt{5}}{3}$

27. $\frac{1-\lambda}{\sqrt{(1-\lambda)^2 + (1-\lambda)^2 + 4}} = \cos 60$

$\Rightarrow 4(1-\lambda)^2 = 2(1+\lambda^2) + 4$

$\Rightarrow \lambda^2 - 4\lambda - 1 = 0$

$\lambda = \frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$

$\therefore \lambda = 2 + \sqrt{5}$ or $2 - \sqrt{5}$.

28. If l, m, n are the direction ratios of the line then $l + 2m - n = 0, l + 3m - n = 0$

$\Rightarrow \frac{l}{1} = \frac{m}{0} = \frac{n}{1}$

$\Rightarrow$ line makes 90° with y-axis or is parallel to xz plane

$\Rightarrow$ (C) is correct.

29. Let the equation of the variable plane be $lx + my + nz = p$

$\therefore$ Coordinates of the vertices of tetrahedron is

$(0, 0, 0); \left(\frac{p}{l}, 0, 0\right); \left(0, \frac{p}{m}, 0\right)$ and

$\left(0, 0, \frac{p}{n}\right)$. If (x_1, y_1, z_1) be the centroid of tetrahedron, then

$x_1 = \frac{p}{4l}, y_1 = \frac{p}{4m}, z_1 = \frac{p}{4n}$

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$$\therefore \frac{p}{l} = 4x, \frac{p}{m} = 4y_1, \frac{p}{n} = 4z_1$$

Also, volume of tetrahedron = $64k^3$

$$\text{i.e., } \frac{1}{6} \begin{vmatrix} \frac{p}{l} & 0 & 0 \\ 0 & \frac{p}{m} & 0 \\ 0 & 0 & \frac{p}{n} \end{vmatrix} = 64k^3$$

$$\Rightarrow \frac{1}{6} \begin{vmatrix} 4x_1 & 0 & 0 \\ 0 & 4y_1 & 0 \\ 0 & 0 & 4z_1 \end{vmatrix} = 64k^3$$

$$\Rightarrow \frac{64x_1y_1z_1}{6} = 64k^3$$

$$\Rightarrow x_1y_1z_1 = 6k^3$$

$\therefore$ Locus of centroid is $xyz = 6k^3$

30. If l, m, n are the direction ratios of line of intersection, then $\ell + 2m + 3n = 0, 3\ell + 3m + n = 0$

$$\Rightarrow \frac{\ell}{1} = \frac{m}{2} = \frac{n}{2}$$

Now y-axis has direction ratios 0, 1, 0

$$\Rightarrow \cos \theta = \frac{2}{\sqrt{6}} \Rightarrow \cos 2\theta = 2 \times \frac{4}{6} - 1 = \frac{1}{3}$$

$$\Rightarrow \sec 2\theta = 3 \Rightarrow \theta = \frac{1}{2} \sec^{-1} 3$$

31. $p_1 = \frac{4}{\sqrt{29}}, p_2 = \frac{1/2}{\sqrt{29}} = \frac{1}{2\sqrt{29}}, p_3 = \frac{8}{\sqrt{29}}$

For these values all the choices are easily verified

32. The given lines are perpendicular for all values of λ and μ if the vectors

$2\hat{i} + p\hat{j} + 5\hat{k}$ and $3\hat{i} - p\hat{j} + p\hat{k}$ are perpendicular.

$$\Rightarrow 2 \times 3 + p(-p) + 5p = 0$$

$$\Rightarrow p^2 - 5p - 6 = 0 \Rightarrow p = -1 \text{ or } 6$$

33. The given lines can be written as

$$\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-2}{0} \quad \dots(1)$$

$$\text{and } \frac{x-1}{1} = \frac{y+\frac{3}{2}}{3/2} = \frac{z+5}{2} \quad \dots(2)$$

Direction numbers of the two lines are

respectively $(3, -2, 0)$ and $\left(1, \frac{3}{2}, 2\right)$

Since $3 \times 1 + (-2) \times \frac{3}{2} + 2 \times 0 = 0$, the given lines are at right angles.

34. Equation of bisector planes are

$$\frac{2x-3y+6z+8}{\sqrt{2^2+3^2+6^2}} = \pm \frac{x-2y+2z+5}{\sqrt{(1)^2+(-2)^2+2^2}}$$

$$\Rightarrow 13x - 23y + 32z + 59 = 0 \text{ \& } x - 5y - 4z + 11 = 0$$

35. Normal to plane P_1 is

$$\vec{n}_1 = (2\hat{j} + 3\hat{k}) \times (4\hat{j} - 3\hat{k}) = -18\hat{i}$$

normal to plane P_2 is

$$\vec{n}_2 = (\hat{j} - \hat{k}) \times (3\hat{i} + 3\hat{j}) = 3\hat{i} - 3\hat{j} - 3\hat{k}$$

$$\therefore \vec{A} \text{ is parallel to } \pm (\vec{n}_1 \times \vec{n}_2) = \pm (-54\hat{j} + 54\hat{k})$$

$\therefore$ angle between $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$ is

$$\cos \theta = \pm \frac{(-54\hat{j} + 54\hat{k}) \cdot (2\hat{i} + \hat{j} - 2\hat{k})}{54\sqrt{2} \cdot 3} = \pm \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

36. Any plane through the second line is $2x + y + z - 1 + k(3x + y + 2z - 2) = 0$

If this is parallel to the first line, then $(2+3k) + (1+k) + (1+2k) = 0 \Rightarrow k = -\frac{2}{3}$

$$\Rightarrow \text{Plane is } 2x + y + z - 1 - \frac{2}{3}(3x + y + 2z - 2) = 0$$

or $y - z + 1 = 0$. The required SD must be distance of this plane from any point on the line $x = y = z$ say $(1, 1, 1)$

$$\Rightarrow \text{SD} = \frac{|1-1+1|}{\sqrt{0^2+1^2+(-1)^2}} = \frac{1}{\sqrt{2}}$$

37. In both systems the $\perp$ distance of plane

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ or $\frac{x}{a'} + \frac{y}{b'} + \frac{z}{c'} = 1$ from origin will be same

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$$\Rightarrow \left| \frac{-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} \right| = \left| \frac{-1}{\sqrt{\frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}}} \right| \text{ etc}$$

38. The line l is perpendicular to l_1 and l_2 . Hence the direction ratios of l are given by the vector

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = -2\hat{i} + 3\hat{j} - 2\hat{k}$$

Let $P \equiv (2r, -3r, 2r)$

As it lies in l_1 , we have

$$\frac{2r-3}{1} = \frac{-3r+1}{2} = \frac{2r-4}{2}$$

Thus we have $r=1$ and $P \equiv (2, -3, 2)$

A point on l_2 is $(3+2s, 3+2s, 2+s)$ then

$$(3+2s-2)^2 + (3+2s+3)^2 + (2+s-2)^2 = 17$$

$$\Rightarrow 9s^2 + 28s + 20 = 0 \text{ given } s = -2, -10/9$$

Thus the points are $(-1, 1, 0)$ and $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$

39. Coefficient det = $\begin{vmatrix} 2 & 5 & 3 \\ 1 & -1 & 4 \\ 0 & 7 & -5 \end{vmatrix} = -46 + 25 + 21 = 0$

So the planes will meet on a line.

for option (c) the distance of the point $(1, 1, 1)$

to the plane $x - y + 4z = 2$ is $\frac{\sqrt{2}}{3}$

40. Let A, B, C be $(\alpha, 0, 0)$, $(0, \beta, 0)$ and $(0, 0, \gamma)$ then

the plane ABC is $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$

Since it always passes through a, b, c

$$\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 1$$

If p is (u, v, w) then $OP^2 = AP^2 = BP^2 = CP^2$

$$\Rightarrow u^2 + v^2 + w^2 = (u-\alpha)^2 + v^2 + w^2 = \dots$$

$$\Rightarrow \alpha = \frac{u}{2}, \beta = \frac{v}{2}, \gamma = \frac{w}{2}$$

On putting α, β, γ in (1) we get

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$$\frac{a}{u} + \frac{b}{v} + \frac{c}{w} = 2 \Rightarrow \text{locus of } (u, v, w) \text{ is}$$

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$$

41) $\frac{x-5}{0} = \frac{y-0}{3-\alpha} = \frac{z-0}{2}$ and

$$\frac{x-\alpha}{0} = \frac{y-0}{-1} = \frac{z-0}{2-\alpha}$$

are the given lines. Then line will be coplaner if

$$\begin{vmatrix} 5-\alpha & 0 & 0 \\ 0 & 3-\alpha & -2 \\ 0 & -1 & 2-\alpha \end{vmatrix} = 0$$

$$\Rightarrow (5-\alpha)\{(3-\alpha)(2-\alpha)-2\} = 0$$

$$\Rightarrow (5-\alpha)(\alpha^2 - 5\alpha + 4) = 0$$

COMPREHENSION QUESTIONS

42. The DC's of the lines are given by $ul + vm + wn = 0$ and Eliminating n between them, we get

$$al^2 + bm^2 + c \left\{ -\frac{(ul + vm)}{w} \right\}^2 = 0$$

$$\Rightarrow (aw^2 + cu^2)l^2 + (bw^2 + cv^2)m^2 + 2cuvlm = 0$$

$$\Rightarrow (aw^2 + cu^2) \left(\frac{l}{m} \right)^2 + 2cuv \left(\frac{l}{m} \right)$$

$$+ (bw^2 + cv^2) = 0 \dots (i)$$

43. Put $u = v = w = 1$ in equation (i), then

$$(a+c) \left(\frac{l}{m} \right)^2 + 2c \left(\frac{l}{m} \right) + (b+c) = 0$$

$$\text{Similarly, } (a+b) \left(\frac{m}{n} \right)^2 + 2a \left(\frac{m}{n} \right) + (c+a) = 0$$

$$\text{and } (b+c) \left(\frac{n}{l} \right)^2 + 2b \left(\frac{n}{l} \right) + (a+b) = 0 \dots (ii)$$

$$\text{From Eq. (ii) } \frac{n_1}{l_1} \cdot \frac{n_2}{l_2} = \frac{(a+b)}{(b+c)}$$

Similarly, $\frac{l_1 l_2}{(b+c)} = \frac{m_1 m_2}{(c+a)} = \frac{n_1 n_2}{(a+b)} \dots (iii)$

$$\therefore \frac{m_1 m_2}{l_1 l_2} = \frac{(c+a)}{(b+c)}$$

44. From Eq. (iii), $\frac{l_1 l_2}{(b+c)} = \frac{m_1 m_2}{(c+a)} = \frac{n_1 n_2}{(a+b)}$

$$= \frac{l_1 l_2 + m_1 m_2 + n_1 n_2}{(b+c) + (c+a) + (a+b)}$$

(by law of proportion)

$$\therefore \text{Lines are perpendicular } l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

Then, $(b+c) + (c+a) + (a+b)$ must be zero

$$\therefore 2a + 2b + 2c = 0$$

$$\Rightarrow a + b + c = 0$$

45. From Eq. (i),

$$(aw^2 + cu^2) \left(\frac{l}{m} \right)^2 + 2cuv \left(\frac{l}{m} \right) + (bw^2 + cv^2) = 0$$

Since, the given lines are parallel, then DC's are equal, then the roots of Eq. (i) are equal.

using discriminant = 0

$$\Rightarrow (2cuv)^2 = 4(aw^2 + cu^2)(bw^2 + cv^2)$$

$$\Rightarrow 4c^2 u^2 v^2 = 4(abw^4 + acv^2 w^2 + bcu^2 w^2 + c^2 u^2 v^2)$$

$$\Rightarrow abw^2 + acv^2 + bcu^2 = 0$$

$$\text{or } \frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = 0$$

$$\Rightarrow \sum \frac{u^2}{a} = 0$$

46. Equation $\vec{r} = \vec{a} + t\hat{n}$ is line passing through $\vec{a}$ and parallel to $\hat{n}$. This will meet the plane $\vec{r} \cdot \hat{n} = d$ at point for which $(\vec{a} + t\hat{n}) \cdot \hat{n} = d \Rightarrow t = d - \vec{a} \cdot \hat{n}$
Required distance =

$$\left| (\vec{a} + (d - \vec{a} \cdot \hat{n})\hat{n}) - \vec{a} \right|$$

$$= |d - \vec{a} \cdot \hat{n}|$$

47. Foot of the perpendicular from the point A to plane $\vec{r} \cdot \hat{n} = d$

$$= \vec{a} + (d - \vec{a} \cdot \hat{n})\hat{n}$$

48. Let $\vec{b}$ be position vector of image of $\vec{a}$

$$\therefore \frac{\vec{b} + \vec{a}}{2} = \vec{a} + (d - \vec{a} \cdot \hat{n})\hat{n}$$

$$\vec{b} = \vec{a} + 2(d - \vec{a} \cdot \hat{n})\hat{n}$$

49. The equation of any plane through the intersection of P_1 and P_2 is

$$P_1 + \lambda P_2 = 0$$

$$\Rightarrow (2x - y + z - 2) + \lambda(x + 2y - z - 3) = 0$$

.....(i)

Since it passes through (3, 2, 1), then $(6 - 2 + 1 - 2) + \lambda(3 + 4 - 1 - 3) = 0$

$$\therefore \lambda = -1$$

From Eq. (i) $x - 3y + 2z + 1 = 0$

which is the required plane.

50. The equation of any plane through (-1, 3, 2) is $a(x+1) + b(y-3) + c(z-2) = 0$ (ii)

If this plane (ii) is perpendicular to P_1 , then

$$2a - b + c = 0 \dots (ii)$$

and if the plane (ii) is perpendicular to P_2 then

$$a + 2b - c = 0 \dots (iii)$$

From Eqs. (ii) and (iii), we get $\frac{a}{-1} = \frac{b}{3} = \frac{c}{5}$

Substituting these proportionate values of a, b, c in Eq. (ii), we get the required equation as

$$-(x+1) + 3(y-3) + 5(z-2) = 0$$

$$\text{or } x - 3y - 5z + 20 = 0.$$

51. The given planes can be written as $-2x + y - z + 2 = 0$ and $-x - 2y + z + 3 = 0$

here, $(-2)(-1) + (1)(-2) + (-1)(1) = -1 < 0$

Equation of bisectors

$$\frac{(-2x + y - z + 2)}{\sqrt{(4+1+1)}} = \pm \frac{(-x - 2y + z + 3)}{\sqrt{1+4+1}}$$

$\therefore$ Acute angle bisector is

$$(-2x + y - z + 2) = (-x - 2y + z + 3)$$

$$\Rightarrow x - 3y + 2z + 1 = 0$$

52. Equation of bisector which not containing the origin

$$\frac{(-2x + y - z + 2)}{\sqrt{(4+1+1)}} = -\frac{(-x - 2y + z + 3)}{\sqrt{(1+4+1)}}$$

$$\Rightarrow 3x + y - 5 = 0$$

53. The image of plane P_1 in the plane mirror P_2 ,

$$\text{then } 2(2 \cdot 1 + (-1)2 + 1(-1))(x + 2y - z - 3)$$

$$= (1 + 4 + 1)(2x - y + z - 2)$$

$$\Rightarrow -(x + 2y - z - 3) = 3(2x - y + z - 2)$$

$$\Rightarrow 7x - y + 4z - 9 = 0$$

54. (A) Let l, m, n be the direction cosines of line of shortest distance.

$$\text{Then } l + 2m + 3n = 0, 2l + 3m + 4n = 0 \text{ and}$$

$$\frac{l}{-1} = \frac{-m}{-2} = \frac{n}{-1} = k \text{ Direction cosines are}$$

$$\left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right) \text{ Shortest distance} =$$

$$\left| -\frac{1}{\sqrt{6}}(1) + 2\left(\frac{2}{\sqrt{6}}\right) - \frac{1}{\sqrt{6}}(4) \right| = \frac{1}{\sqrt{6}}$$

55. (C) $P(1, 0, 0), O(0, 0, 0), Q(2, -2, -1)$

$$\Delta_x = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ -2 & -1 & 1 \end{vmatrix}, \Delta_y = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 2 & -1 & 1 \end{vmatrix}, \Delta_z = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 2 & -2 & 1 \end{vmatrix}$$

$$\Delta^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2 \Rightarrow \Delta = \frac{\sqrt{5}}{2}$$

56. (A) With help of two equations

$$\left(\frac{l}{m}\right)^2 - 2a\left(\frac{l}{m}\right) + 1 - a^2 = 0$$

$$\text{For parallel lines } D = 0$$

$$\Rightarrow 4a^2 - 4(1 - a^2) = 0$$

$$\Rightarrow a = \pm \frac{1}{\sqrt{2}}$$

57. Suppose the 3 plane intersect in line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$

$$\Rightarrow -l + m \sin m\psi + n \sin f = \phi = 0$$

$$l \sin \psi - m + n \sin \phi = 0$$

$$l \sin \phi + m \sin \theta - n = 0$$

$$\Rightarrow \begin{vmatrix} -1 & \sin \psi & \sin \phi \\ \sin \psi & -1 & \sin \theta \\ \sin \phi & \sin \theta & -1 \end{vmatrix}$$

$$\Rightarrow \sin^2 \theta + \sin^2 \psi + \sin^2 \phi + 2 \sin \theta \sin \phi \sin \psi = 1$$

58. $\sin^2 \theta + \sin \theta (2 \sin \phi \sin \psi) + (\sin^2 \psi \sin^2 \phi - 1) = 0$

$$\Rightarrow \sin \theta = -\sin \phi \sin \psi \pm \sqrt{\sin^2 \phi \sin^2 \psi - (\sin^2 \phi + \sin^2 \psi - 1)}$$

$$\Rightarrow -\sin \psi \sin \phi \pm \cos \phi \cos \psi$$

$$\Rightarrow -\cos(\phi + \Psi)$$

$$\therefore \theta + \phi + \Psi = \frac{\pi}{2}$$

59. From 1

$$\frac{l}{\cos^2 \theta} = \frac{m}{\sin \theta \sin \phi + \sin \Psi} = \frac{n}{\sin \theta \sin \Psi + \sin \phi}$$

$$\therefore \frac{l}{\cos \theta} = \frac{m}{\cos \phi} = \frac{n}{\cos \Psi}$$

PASSAGE - VI

60. Let $P = (\alpha, \beta, \gamma)$

$$\Rightarrow \frac{a}{a} + \frac{\beta}{\beta} + \frac{\gamma}{c} = 1$$

$$\text{Dr's of OP } \alpha, \beta, \gamma$$

$\therefore$ Equation of plane passing through P and perpendicular to OP is

$$\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0$$

$$\Rightarrow ax + \beta y + \gamma z = \alpha^2 + \beta^2 + \gamma^2$$

$$\Rightarrow A = \left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, 0, 0 \right)$$

61. Equation of planes through A, B, C are

$$x = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, \gamma \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}$$

$$\text{and } z = \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta} \text{ respectively}$$

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$$\therefore \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{\alpha^2 + \beta^2 + \gamma^2} \quad (1)$$

$$\text{and } \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} = \frac{\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c}}{\alpha^2 + \beta^2 + \gamma^2} \quad (2)$$

$$\text{From 1 and 2 } \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} = \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz}$$

PASSAGE - VII

62. M&N must be (a,0,c) and (a,b,c) respectively. Now equation of plane OMN can

$$\text{be easily found as } \frac{x}{a} - \frac{y}{b} - \frac{z}{c} = 0$$

63. Dir's of OP=a,b,c

$$\text{Dir's of normal to OMN} = \frac{1}{a}, -\frac{1}{b}, -\frac{1}{c}$$

$$\Rightarrow \sin \theta = \frac{a \cdot \frac{1}{a} - b \cdot \frac{1}{b} - c \cdot \frac{1}{c}}{\sqrt{a^2 + b^2 + c^2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$= \frac{abc}{\sqrt{a^2 + b^2 + c^2} \sqrt{a^2 b^2 + b^2 c^2 + c^2 a^2}}$$

64. α be the angle which OP makes with yz plane or with x=0

$$\Rightarrow \sin \alpha = a \cdot 1 + b \cdot 0 + c \cdot 0$$

$$\Rightarrow \sin \alpha = \frac{a \cdot 1 + b \cdot 0 + c \cdot 0}{\sqrt{1^2 + 0^2 + 0^2} \sqrt{a^2 + b^2 + c^2}} = \frac{a}{\sqrt{a^2 + b^2 + c^2}}$$

65. Given lines $L_1 : \frac{x-1}{-2} = \frac{y}{1} = \frac{z+1}{1}$

$$L_2 : \frac{x-4}{1} = \frac{y-5}{4} = \frac{z+2}{-1}$$

Angle θ between the lines is given by

$$\cos \theta = \frac{1}{6\sqrt{3}}$$

66. Let the required plane π be

$$4x - y - 11 + \lambda(x + z - 2) = 0$$

$$\pi \text{ is parallel to } L_1 \Rightarrow \lambda = -9$$

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67. Distance of the plane π from the line L_1 = the distance from (1,0,-1) to the plane $\pi = \frac{11}{\sqrt{107}}$

$$68. \vec{n}_1 = 3\hat{i} - 3\hat{j} + \hat{k} \text{ and } \vec{n}_2 = \hat{i} - \hat{j} + c\hat{k}$$

$$\text{Now } \cos 45^\circ = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} = \frac{|3 + 3 + c|}{\sqrt{19} \sqrt{2 + c^2}}$$

$$\therefore 2(6 + c)^2 = 19(2 + c^2)$$

$$2(36 + c^2 + 12c) = 38 + 19c^2$$

$$17c^2 - 24c - 34 = 0$$

$$\text{Product of roots} = -2$$

$$69. x = k + 2; y = 3k + 1 \text{ and } z = -k + 5$$

They must satisfy then equation of plane A

$$3(k + 2) - 3(3k + 1) + (-k + 5) = 2$$

$$\Rightarrow k = -\frac{6}{7}$$

$$\text{Hence M is the point } \left(\frac{8}{7}, -\frac{11}{7}, \frac{41}{7} \right)$$

70. Plane P_1 is of the form

$$\vec{r} \cdot \vec{n}_1 = 0, \text{ where } \vec{n}_1 = (4, -3, 7)$$

Plane P_2 is of the form $\vec{r} \cdot \vec{n}_2 = 0$, where $\vec{n}_2 = (2, 1, -5)$

The vector $\vec{b}$ along the line of intersection of planes is $\vec{n}_1 \times \vec{n}_2 = (4, 17, 5) = \vec{n}_3$

Since the line of greatest slope is perpendicular to $\vec{n}_3$ and $\vec{n}_2$ the vector along the line of greatest slope $= \vec{n}_2 \times \vec{n}_3 = (3, -1, 1) = \vec{n}_4$

$$\text{and the unit vector } \hat{n}_4 = \left(\frac{3}{\sqrt{11}}, \frac{-1}{\sqrt{11}}, \frac{1}{\sqrt{11}} \right)$$

71. The point on the line $\frac{x}{3} = \frac{y}{-1} = \frac{z}{1}$ at a distance $2\sqrt{11}$

$$\text{units from the origin is given by } \frac{x}{\frac{3}{\sqrt{11}}} = \frac{y}{\frac{-1}{\sqrt{11}}} = \frac{z}{\frac{1}{\sqrt{11}}} = 2\sqrt{11}$$

$$\text{The point is } (6, -2, 2)$$

72. The circumcentre will lie in the plane ABC

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whose equation is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

73. If (x, y, z) is the circumcentre, then

$$(xa)^2 + y^2 + z^2 = x^2 + (yb)^2 + z^2 = x^2 + y^2 + (zc)^2$$

$ax by = a^2b^2$ and $by cz = b^2c^2$

$$x = \frac{by + a^2 - b^2}{a} \text{ and } z = \frac{by + c^2 - b^2}{c}$$

on putting in $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, We get

$$\frac{by + a^2 - b^2}{a^2} + \frac{y}{b} + \frac{by + c^2 - b^2}{c^2} = 1$$

$$y = \left(\frac{b^2c^2 + a^2b^2 - a^2c^2}{a^2b^2 + b^2c^2 + c^2a^2} \right) b(C) \text{ is correct}$$

74. On applying $\frac{OG}{GH} = \frac{1}{2}$ and using the fact that G

has coordinates $\frac{a}{3}, \frac{b}{3}, \frac{c}{3}$ and easily get (A)

as correct

75. It is obvious

76. Statement 1:

$$\vec{r} \times (2\hat{i} - \hat{j} + 3\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 2 & -1 & 3 \end{vmatrix}$$

$$= \hat{i}(3y + z) - \hat{j}(3x - 2z) + \hat{k}(-x - 2y)$$

$$= 3\hat{i} + \hat{k}$$

$$3y + z = 3, 3x - 2z = 0, -x - 2y = 1$$

$$\therefore 3x + 6y = 6 \Rightarrow x + 2y = 2 \quad \dots(i)$$

$$x + 2y = -1 \quad \dots(ii)$$

(i) and (ii) are parallel planes not straight line
Statement-2

$$\vec{r} \times (\hat{i} + 2\hat{j} - 3\hat{k})$$

$$= \hat{i}(-3y - 2z) - \hat{j}(3x - z) + \hat{k}(2x - y)$$

$$= 2\hat{i} - \hat{j}$$

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i.e., $-3y - 2z = 2, 3x + z = -1, 2x - y = 0$

These are intersecting in a line

77. Conceptual

78. Here, $a = 2, b = -6,$

$c = 4, f = 9, g = 1,$

$$\backslash abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= (2)(-6)(4) + 2(9)(1) \left(\frac{1}{2} \right)$$

$$- (2)(81) + 6(1)^2 - 4 \left(\frac{1}{2} \right)^2$$

$$= -48 + 9 - 162 + 6 - 1$$

$$= -196 \neq 0$$

$\therefore$ Given equation cannot represent a pair of planes.

79. (B)

Let l, m, n be the DC's of the line of the common perpendicular (or SD) to the two given lines.

Then, we have $-4l + 3m + 2n = 0$

$$\text{and } -4l + 1m + 1n = 0$$

Solving these, we get

$$\frac{l}{3-2} = \frac{m}{-8+4} = \frac{n}{-4+12} \text{ or}$$

$$\frac{l}{1} = \frac{m}{-4} = \frac{n}{8} = \frac{\sqrt{(\ell^2 + m^2 + n^2)}}{\sqrt{(1)^2 + (-4)^2 + (8)^2}} = \frac{1}{9}$$

$$\therefore \text{DC's of SD are } \frac{1}{9}, \frac{4}{-9}, \frac{8}{9}$$

Also, $A(-3, 6, 0)$ is a point on first line and

$B(-2, 0, 7)$ is a point on second line, then

$$\text{SD} = \left| (-2+3)\frac{1}{9} + (0-6)\left(-\frac{4}{9}\right) + (7-0)\left(\frac{8}{9}\right) \right|$$

$$= 9$$

and two lines are said to be skew lines or non intersecting lines if they do not lie in the same plane.

80. $x + y + z - 6 + \lambda(2x + 3y + 4z + 5) = 0$

which passes through $(4, 4, 4)$

$$\Rightarrow 29x + 23y + 17z = 276$$

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81. Dr's of $L_1 : 0, 1, 1$

Dr's of $L_2 : 0, 1, 1$

Dr's of $L_3 : 0, 1, 1$

hence statement 1 is false

82. (A) $1(t) + 2(2t+1) + 2(2t) = 0 \Rightarrow t = -\frac{9}{2}$

(B) $a(x-1) + b(x+2) + 2(x-1) = 0$

and $2a - 2b + c = 0, a - b + 2c = 0$

(C) $\frac{1}{6} [\overline{AB} \overline{AC} \overline{AD}]$

(D) Use $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$

83. (A) $\frac{t(\overline{a} \times \overline{b}) \cdot \overline{c}}{|\overline{c}|}$

(B) $\lambda(\overline{n}_1 \times \overline{n}_2)$

(C) $a\overline{i} + c\overline{k} = 0, a\overline{i} + b\overline{j} = 0$

(D) by verification

84. Use the concept of coplanarity of lines and planes in 3-dimensional space

85. (A) Take any point on L_2 and find the parameter to get the perpendicular distance

(B) $[\overline{c} - \overline{a} \quad \overline{b} \quad \overline{d}]$

86. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

87. Equation of line (1) is

$$\frac{x-5}{-2} = \frac{y+2}{0} = \frac{z}{1}$$

Equation of line (2) is

$$\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$$

$$\text{Shortest distance} = \frac{|(\overline{a} - \overline{c}) \cdot \overline{b} \times \overline{d}|}{|\overline{b} \times \overline{d}|} = \frac{4}{2} = 2 \text{ units}$$

88. Let $P = (x, y, z)$ then $OP^2 = 4 + 9t^2 \geq 4$

89. The system of planes has no solution

$$\begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & 2 \\ 3 & \lambda & 1 \end{vmatrix} = 0 \Rightarrow (\lambda - 4)(\lambda + 3) = 0$$

$$\therefore \lambda = 4$$

90. Let the equation of required plane be

$$2x - y + z - 3 + \lambda(3x + y + z - 5) = 0$$

Distance of this plane from $(2, 1, -1)$ is $\frac{1}{\sqrt{6}}$

$$\Rightarrow \lambda = \frac{-24}{5}. \text{ The equation of required plane is}$$

$$62x + 29y + 19z + 135 = 0$$

91. Let a, b, c be the D.r's of normal to the plane
Given $a - c = 0, -a + b = 0$

D.r's of normal to the plane are $(1, 1, 1)$ equation of the plane is $x + y + z = 3$ now

$$A = (3, 0, 0), B = (0, 3, 0), C = (0, 0, 3)$$

$$V = \frac{1}{6} \begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix} \Rightarrow 2V = 9$$

$$92. \left(\frac{a+3}{4}, \frac{b+3}{4}, \frac{5+c}{4} \right) = (1, 2, 3)$$

$$a = 1, b = 5, c = 7$$

93. Area of the triangle ABC is

$$\Delta = \frac{1}{2} \begin{vmatrix} i & j & k \\ 2 & -2 & 0 \\ 2 & 0 & -2 \end{vmatrix} = 2\sqrt{3}$$

equation of the plane ABC is $x + y + z - 1 = 0$

angle α between plane $x + y + z - 1 = 0$ and the xy

plane is given by $\cos \alpha = \frac{1}{\sqrt{3}}$

Then $M = \Delta \cos \alpha = 2$

$$\therefore \frac{m(m+1)}{2} = 3$$

94. Use $SA = SB = SC$ with $(-1, \lambda, -3)$

95. Let equation of new plane is
 $2x - 2y + z - 3 + \lambda z = 0$ point $(3, 1, 1)$ lies on it
 $\Rightarrow \lambda = -2$

hence the new plane is $2x - 2y - z = 3$

$$\text{Now } \cos \alpha = \frac{4+4-1}{3.3} = \frac{7}{9}$$